SOFTWARE ENGINEERING LAB

Course Code: CSE3132

Test Cases: Hackathon Project Team Formation System (TeamSpark)

Submitted by: Bhavya Bansal (2427030421), Raj Shukla (2427030683)

Date: 20 October 2026

1. Purpose

This document lists the test cases written for TeamSpark and the result of running them. Each test checks one requirement from our SRS: we give the system a known input and compare what it does with what the SRS says it should do.

2. Testing Tools and Approach

- Jest (logic testing, 13 tests, TC-01 to TC-13): runs the real application code from teamspark.html and checks the Compatibility Score formula, search, suggestions, access rules, requests, listings and data storage.
- Playwright (browser testing, 7 tests, UI-01 to UI-07): opens TeamSpark in a real Chromium browser, types and clicks like a user, and checks what appears on the screen.
- Why two tools: Jest proves the logic is right; Playwright proves the same features work through the screen. Selenium and Cypress are alternatives to Playwright that automate a browser in the same way; Playwright was chosen because it needs no separate driver and its tests are short and reliable.
- Each test starts from a fresh copy of the application with the same sample data, so tests do not affect one another.

3. Test Environment

- Node.js, Jest 29.7, Playwright 1.56 on Chromium. Application under test: teamspark.html. Test files: teamspark.test.js and e2e/teamspark.spec.js.
- Sample data: 4 listings, 6 sample students, 1 pending request, current user "You".
- Run: npm install, then npm run test:all.

4. Result Summary

Total test cases: 20 (13 Jest + 7 Playwright). Passed: 20. Failed: 0.

5. Test Case Table

Test ID	SRS requirement	Module	What is done (input)	Expected result	Status
TC-01	REQ-TS-07	Scoring	skillMatch when student has 2 of the 3 required skills	0.6667	Pass
TC-02	REQ-TS-07	Scoring	workloadCap for 0, 1, 3 and 5 current projects	1.0, 0.6667, 0 and 0 (never negative)	Pass
TC-03	REQ-TS-07	Scoring	total Compatibility Score of Arjun Rao for "Smart Attendance App"	0.8933	Pass
TC-04	REQ-TS-06	Search	search "python", then " REACT " (capitals and spaces)	1 match for python, 3 matches for react	Pass
TC-05	REQ-TS-08	Suggestions	findMatches for "Smart Attendance App"	top 3 are Arjun, Kavya, Aditi in this order; others not shown	Pass
TC-06	REQ-TS-08, 09	Access rules	open own listing, then someone else's listing	owner sees "Find suggested candidates" only; other user sees "Apply to join" only	Pass
TC-07	REQ-TS-09, 10	Join request	apply to a listing, then owner accepts	request is Pending first, then Accepted and the listing is added to memberOf	Pass
TC-08	REQ-TS-04	Create listing	post a valid listing, then post one with an empty title	valid one is added first with owner "You" and lowercase skills; empty title is rejected with an error	Pass
TC-09	REQ-TS-05	Delete listing	delete a listing that has a join request	listing and its request are both removed	Pass
TC-10	NFR-TS-05	Data storage	create a listing and reload; then load corrupted stored data	listing is restored after reload; corrupted data falls back to the 4 sample listings without crashing	Pass
TC-11	REQ-TS-03	Profile	save skill levels 0, 1, 5, 6 and "abc" (boundary values)	levels become 1, 1, 5, 5 and 3 (kept inside 1 to 5; non-number defaults to 3)	Pass
TC-12	REQ-TS-04	Create listing	post a listing that has a title but no skills	rejected with an error, nothing added	Pass
TC-13	REQ-TS-10	Join request	reject a request; then accept the same listing twice	rejecting adds no membership; accepting twice keeps one membership (no duplicate)	Pass
UI-01	REQ-TS-06	Search	type "python" in the search box, then type "cobol"	only ML Study Recommender for python; "No listings match" message for cobol	Pass
UI-02	REQ-TS-04	Create listing	fill the form and click Post listing; then submit with an empty title	new listing appears in Browse (5 teams); empty title shows an error and adds nothing	Pass
UI-03	REQ-TS-08, 09	Access rules	open your own listing, then another student's listing	your listing shows only "Find suggested candidates"; the other shows only "Apply to join"	Pass

Test ID	SRS requirement	Module	What is done (input)	Expected result	Status
UI-04	REQ-TS-08	Suggested candidates	post a listing needing react, node.js, mongodb; open it and click Find suggested candidates; invite the first candidate	3 candidates, Arjun Rao first with 89%; Invite button turns into disabled "Invited"	Pass
UI-05	REQ-TS-09	Join request	open Smart Attendance App and click Apply to join; then open the same listing again	Requests page shows the application as Pending; reopening the listing shows a disabled "Request sent" button	Pass
UI-06	REQ-TS-05	Delete listing	click Delete on your own listing twice	first click asks "Click again to confirm"; second click removes the listing (3 teams left)	Pass
UI-07	NFR-TS-09	Mobile layout	open the site at phone width (375 x 700 px)	listings visible and the page does not scroll sideways	Pass

6. Scope and Limitations

- The tests cover the main features of the prototype: scoring and suggestions, search, listings, join requests, access rules, storage and phone layout. Browsers other than Chromium were not run.
- The prototype has no server, login or database, so registration, login, notifications and admin requirements (REQ-TS-01, 02, 11 to 16) are not tested here. They belong to the full implementation.
- Performance, security and accessibility requirements need separate tools and were not part of this round.